

Qualified Sets of Identified 18B20 Probes

Keep four identities stable while evidence changes

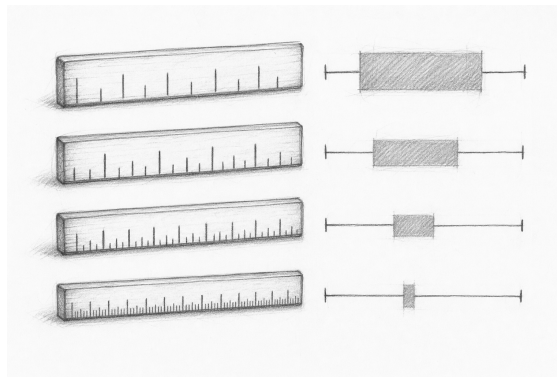
E0 SYNTHETIC REPLAY — NO BUS, PROBE, OR THERMAL CLAIM

Lesson	065 — component	Time	180–240 minutes
Prerequisite	Lesson 064 copied transactions	Board	compile-only Mega replay
Input	copied search/conversion/read evidence	Observation	named snapshot cells

Responsibility. `Qualified18B20ProbeSetPolicy` qualifies one bounded copied cycle against exactly four configured family-0x28 ROM identities. It publishes four stable attributed slots with signed temperature code-space intervals, age, quality, and status. It does not discover, power, sample, or calibrate a physical sensor.

1 Qualify evidence, not a box of parts

Lesson 064 proves that a bounded policy can correlate copied single-wire transaction evidence. Lesson 065 gives some of those transactions DS18B20 meaning: a chained ROM search, a conversion start and completion observation, and a nine-byte scratchpad read. Meaning arrives only after structure, identity, chronology, and CRC checks agree.



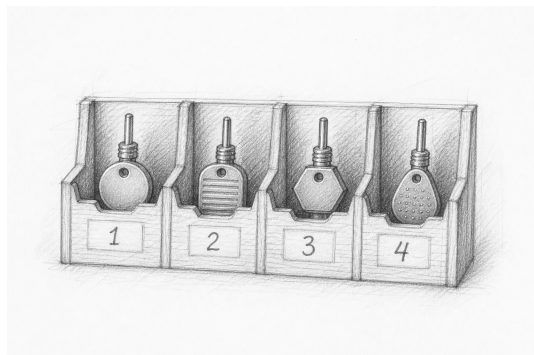
Alternative text: four graphite measuring bars with progressively finer tick marks sit beside four closed intervals that narrow from a broad shaded span to a single narrow band, illustrating resolution without claiming physical accuracy.

E0 can establish	E0 cannot establish
family byte and Dallas ROM CRC	genuine package, maker, or cable identity
copied scratchpad CRC and resolution bits	electrical timing, rise time, or supply current
stable configured-slot attribution	physical position or probe response time
deterministic freshness and step policy	thermal accuracy, waterproofing, or safe use

There is no wiring diagram or formal schematic at E0. A future powered lesson must qualify exact specimens, pin order, external supply, ordinary pull-up, rails, line timing, four-device loading, and independent non-Serial evidence. Parasite power, strong pull-up, immersion, medical, food, preservation, and unattended heat-source claims remain excluded.

2 Configure exactly four stable identities

Configuration contains four distinct complete ROM codes. Each must use family byte `0x28` and pass the Dallas CRC-8 over its first seven bytes. The algorithm begins at zero, processes each byte least-significant bit first, uses reflected polynomial `0x8c`, and applies no final XOR.¹



Alternative text: four numbered graphite storage bins each hold a deliberately different token shaped like a small probe, a visual metaphor for four identities that retain their assigned slots rather than depictions of qualified physical specimens.

Each slot also configures 9-, 10-, 11-, or 12-bit resolution, a conversion deadline, maximum age, inclusive signed raw-sixteenth range, and maximum per-cycle step magnitude. There is no runtime count, empty wildcard, or replacement identity. Search order cannot change configured order.

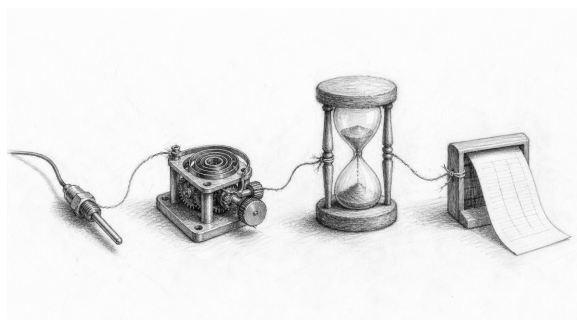
Resolution	Undefined raw bits	Maximum conversion	Represented interval
9 bit	3	94 ms	$[m, m + 7]$ sixteenths
10 bit	2	188 ms	$[m, m + 3]$ sixteenths
11 bit	1	375 ms	$[m, m + 1]$ sixteenths
12 bit	0	750 ms	$[m, m]$ sixteenths

Here m is the signed raw value after masking undefined low bits. The closed interval describes digital code space, not sensor accuracy or uncertainty. The final API requires the configured deadline to equal the resolution maximum for its slot; a shorter or longer value rejects configuration.

1. A 9-bit scratchpad decodes to masked raw value 400. Record the interval: _____ through _____.
2. A search returns the same four ROMs in reverse order. Which configured slot changes identity? _____.
3. Why is an all-zero optional ROM incompatible with this component? _____.

3 Stage one complete cycle

A caller lifetime-reuses one opaque `Ds18B20CycleBuilder`. After `beginCycle()`, role-specific ingestion methods accept one complete Lesson 064 snapshot at a time. `finishSearch()` records whether enumeration completed or explicitly exceeded the four-result evidence capacity. Only `finalizeCycle()` may commit policy state.

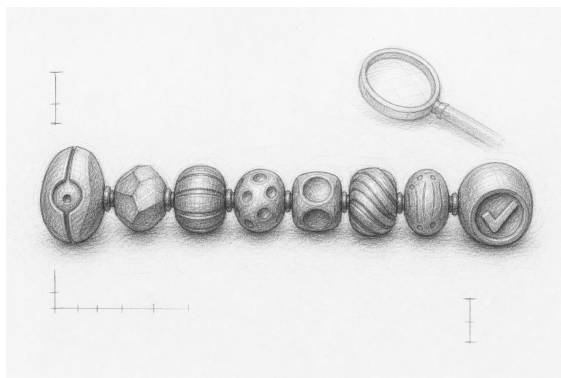


Alternative text: a graphite line runs from a loose probe-shaped token through a bounded geared mechanism, an hourglass, and a paper roll, representing copied input, staged processing, supplied time, and one published result rather than a physical circuit.

¹Analog Devices, *DS18B20 Programmable Resolution 1-Wire Digital Thermometer*, Rev. 6.

4 Make search continuity prove presence

The builder retains at most four `Ds18b20NormalizedSearchPass` records. Each record contains the normalized transaction reference plus its requested and completed `OneWireSearchState`. The first request uses empty search state. Every later request must equal the immediately preceding completed result, including ROM, `lastDiscrepancy`, and `lastDevice`. No pass may follow a terminal result. The request/result chain is the bounded policy rendering of the manufacturer's discrepancy-tree search procedure.²



Alternative text: eight distinct graphite beads form one connected chain, ending in a bead with a check mark; a magnifying glass above suggests inspecting each linked search result and its terminal evidence.

A complete nonempty enumeration ends with `lastDevice` true. One through four returned devices may therefore prove configured identities missing. A zero-pass cycle cannot prove all four missing because it contains no completed search witness. Four results ending nonterminal may explicitly prove over-capacity; the fifth device is never silently dropped.

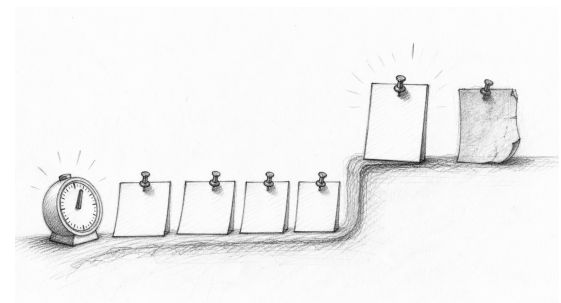
Search evidence	Set result	What it does not imply
complete terminal chain, known distinct ROMs	Complete or Missing	physical location
duplicate returned ROM	DuplicateIdentity	two interchangeable probes
CRC-valid foreign ROM	UnknownIdentity	permission to replace a slot
bad-family or bad-CRC returned ROM	TransportFault	attributable configured identity
truncated, failed, or over-capacity	TransportFault	proved absence

`presentMask` marks configured identities proved present by a structurally complete search. `validCount` counts only **Current** slots. `faultMask` marks every slot that is not **Current**. These fields answer different questions and must not be collapsed.

Predict a complete search that finds three configured ROMs and omits the fourth: set quality _____; missing slot still has its configured ROM? _____; array length _____.

5 Bind conversion and read evidence

For each returned device, conversion start, conversion-status continuation, and scratchpad read must agree on exact ROM, owner, lifecycle, configuration, predecessor transaction, and one nonzero conversion generation. A crossed witness rejects the entire cycle atomically.



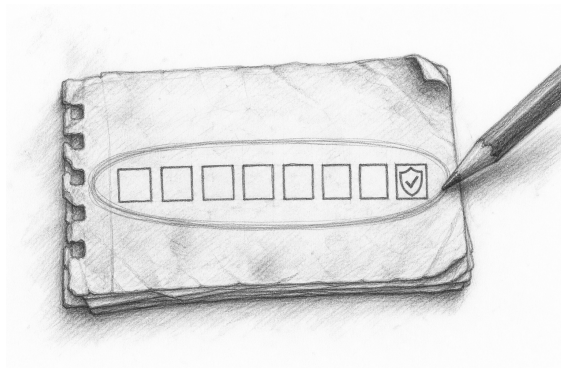
Alternative text: a graphite clock precedes four pinned evidence notes along one path; the path rises to two final notes, one clean and one shaded, illustrating that chronology and correlation determine the admitted outcome.

Completion requires a correlated conversion-status receipt whose copied bit is high. Elapsed time alone does not prove completion. A high bit may arrive before the configured maximum. A low bit exactly at or after that maximum, or a high bit after it, is transport-fault evidence.

²Analog Devices, *1-Wire Search Algorithm*, Application Note 187.

6 Decode scratchpads without inventing precision

A DS18B20 scratchpad is exactly nine bytes. Byte eight must equal the Dallas CRC-8 of bytes zero through seven before temperature or resolution is used. Configuration byte bit 7 must be zero and bits 4-0 must be one; bits 6-5 encode resolution and must match the configured slot.



Alternative text: a pencil points to a graphite notebook strip of ordered byte-like cells whose final cell bears an integrity shield and check mark, emphasizing validation before interpretation.

The published numeric unit remains a signed sixteenth of a degree Celsius. Negative values use the signed 16-bit scratchpad representation. Widened arithmetic forms intervals and absolute step differences so endpoints do not overflow.

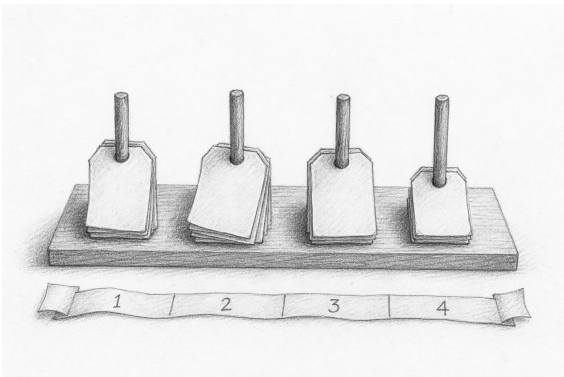
Quality	Interpretation
Current	correlated completion and CRC-valid, configured evidence
ScratchpadCrcFault	nine bytes arrived, but integrity failed
ResolutionMismatch	scratchpad resolution disagrees with its slot
ResetDefaultWithoutConversion	raw +85°C lacks exact completed conversion
ImplausibleStep	widened absolute step exceeds configured maximum
Stale	last trusted value is older than maximum age
Missing	complete search proves configured identity absent
TransportFault	correlated transport/chronology evidence failed

Raw +85°C is not automatically bad. Without exact correlated completion it is the reset-default quality; after correlated completion it is a possible current reading. Likewise, an implausible step is a domain warning, not proof that a physical event was impossible. The new valid sample becomes the next comparison baseline.

1. One scratchpad byte changes but its old CRC is retained. Which numeric fields may refresh? _____.
2. At the maximum configured step, is the new value admitted? _____. One sixteenth beyond? _____.
3. Does a code-space interval authorize an accuracy claim? _____.

7 Preserve trusted values through pending and missing states

Freshness begins at the caller-supplied policy-clock time passed with the correlated scratchpad. Lesson 065 also retains transaction-clock times, but does not invent an epoch mapping between the two clocks. Age equal to `maximumAge` remains current; one millisecond older is stale.



Alternative text: four numbered positions support separate stacks of graphite tags on pegs; the stacks differ in depth while positions one through four remain fixed, illustrating stable identity with retained history.

ConversionPending retains the last trusted raw value, interval, observation time, and increasing age. It does not read an incomplete scratchpad or refresh freshness. **Missing** likewise retains the stable identity and prior trusted value. A later fully correlated current cycle returns that same identity to its original slot.

Set-level structural and transport decisions precede per-slot qualities. A valid cycle commits all four slots together, so one CRC fault does not erase three current readings. Status precedence follows configured slot order after set-level failures; status and quality remain separate facts.

Slot	Search	Evidence	Published quality
0	present	complete, CRC valid	Current
1	present	completion still low, before limit	ConversionPending
2	absent	complete terminal search	Missing
3	present	scratchpad CRC bad	ScratchpadCrcFault

For this table, write `validCount` _____ and the four-bit `faultMask` _____. Which rows may still retain an older trusted numeric value? _____.

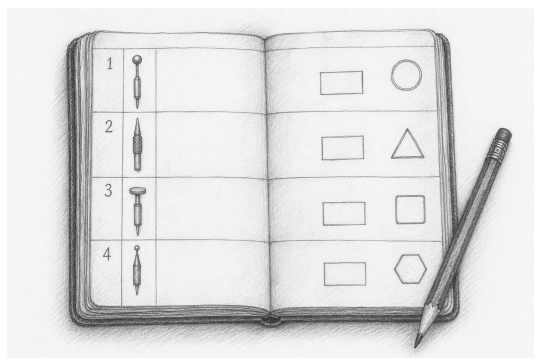
8 Reject structural changes atomically

Cycle sequence is nonzero and modularly forward; `UINT32.MAX` followed by 1 is its defined nonzero rollover. Zero, regression, backward time, and an exact half-range ambiguity reject. An equal sequence accepts only a field-for-field identical completed builder image.

A fieldwise-identical normalized duplicate is idempotent: it cannot refresh age, extend a pending conversion, advance step history, or change disappearance evidence. A changed duplicate rejects without modifying policy state or the caller's output. Ingestion failure leaves every byte of the staged builder unchanged; failed finalization leaves policy and output unchanged.

9 Replay the policy on a Mega without touching hardware

The canonical sketch creates four fixed configurations, replays copied Lesson 064 snapshots into one reusable builder, finalizes one cycle, and copies selected results into named volatile cells. Compile and inspect those cells; do not upload this E0 exercise to claim a powered fixture.



Alternative text: an open graphite notebook has four numbered rows, distinct tool-like identity symbols, blank result rectangles, and different geometric quality marks; a pencil suggests recording the replay outcome without depicting a wired board.

1. **Predict:** record configured slot order, expected qualities, `presentMask`, `faultMask`, and `validCount`.
2. **Observe:** compile for Mega 2560, then inspect the named volatile result cells in a debugger, simulator, or disassembly. Serial is optional supporting context, not the only evidence.
3. **Interpret:** compare each stable ROM slot, quality, raw interval, age, and status with the copied cycle. Explain any disagreement before changing fixture data.

This observation establishes software acquisition and deterministic result publication only. There is no electrical safe-state experiment because the E0 component owns no endpoint. Future E1 resource acquisition and physical released-line evidence are separate checks.

10 Publisher verification and resource boundary

Mega replay	ordinary: 13,662 B flash, 1,438 B static SRAM; isolated no-LTO: 16,196 B flash, 1,438 B static SRAM.
Objects and stack	policy 764 B; reusable builder 477 B; snapshot 180 B; conservative synchronous stack 533 B.
Placement	recurring policy plus builder 1,241 B; lifetime placement including one snapshot 1,421 B; residual SRAM 6,093 B after static, stack, and reserve.
Evidence identity	6e4f9d75a48398919dc764d386cb3c2c 0faf9fb11e9c874c9b0377c768ef0bc3
Publication evidence	Publisher-owned PDF validation and rendered pencil visual review passed; independent resource rereview reproduced the evidence cleanly.
Physical evidence	Deferred; no powered specimen or bus is authorized.

Planning hard limits are 16 KiB ordinary replay flash, 1,536 B static SRAM, 640 B conservative synchronous stack, 768 B policy object, 512 B reusable builder, and 512 B caller-owned snapshot/result. Every hard gate and the residual floor pass. Ordinary flash and SRAM, stack, policy, and builder miss their planning targets but remain below their hard limits; independent review accepted those bounded misses without weakening replay identity or recurring operation. Exact specimen, electrical, and physical E1 acceptance remain open.

11 Exercises and next boundary

1. Construct a complete search permutation that leaves slot order unchanged. Mark which evidence proves one configured ROM missing.
2. Corrupt each scratchpad byte in turn. Confirm that no corruption partially refreshes value, age, presence, or step baseline.
3. Replay pending immediately before, exactly at, and immediately after each resolution deadline. Explain the boundary outcomes.
4. Demonstrate current, pending, stale, missing, and reappeared states for one ROM without ever changing its configured index.
5. Explain why Lesson 066 may use configured ROMs for spatial order but cannot use search order as physical position.

Lesson 066 will consume the complete copied four-slot snapshot and map a configured subset into spatial order, adjacent code-space gradients, fault-dominant presentation pages, and caller-owned record intent. It still will not drive a display, read an RTC, write storage, or turn these digital intervals into calibrated thermal accuracy.

Stop at E0: qualify exact powered specimens and the complete four-device bus before adding wiring, schematics, or physical claims.